

Predicting Medical Insurance Costs with Linear and LASSO Regression

Identifying the key drivers of health insurance charges and building regularised regression models to predict individual medical costs.

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Overview

Medical insurance charges vary widely across individuals and are influenced by demographic and lifestyle factors. This project uses the insurance.csv dataset (1,338 records) to model individual charges from six inputs: age, sex, BMI, number of children, smoker status, and region. Two regression approaches are compared: ordinary linear regression and LASSO (Least Absolute Shrinkage and Selection Operator) regression with cross-validated regularisation.

Goals

Build accurate, interpretable regression models that predict individual medical insurance costs, and determine which input variables have the strongest influence on charges.

Objectives

- Preprocess the dataset: label-encode categorical variables and apply Min-Max scaling to all features.
- Analyse the correlation matrix on the training set to identify the three inputs most correlated with charges (smoker, age, BMI).
- Train a linear regression model on the top three correlated features and evaluate it on training and test sets using MSE.
- Train a LASSO regression model and use LassoCV cross-validation to select the optimal regularisation strength.
- Plot LASSO coefficient paths against the regularisation parameter and report test-set MSE and R-squared for the best model.

Approach

Data is split 70/30 into training and test sets. Categorical variables are label-encoded before scaling. Correlation analysis on the training set guides feature selection for the linear model. The LASSO model is trained on all features, and LassoCV with cross-validation identifies the best lambda value. Coefficient shrinkage plots visualise how regularisation reduces less important features toward zero.

Outcomes

- Smoker status is the single most important predictor of insurance charges, confirmed by both correlation analysis and LASSO coefficients.
- The LASSO model achieves a test-set MSE of approximately 0.0086 and an R-squared of approximately 0.77.

- LASSO regularisation successfully identifies and down-weights less informative features, producing a sparser and more interpretable model than ordinary linear regression.
- Coefficient path plots confirm that smoker status retains a large coefficient across all regularisation strengths while other features shrink toward zero.